POWER SYSTEM DYNAMIC SECURITY ASSESSMENT



October 2026

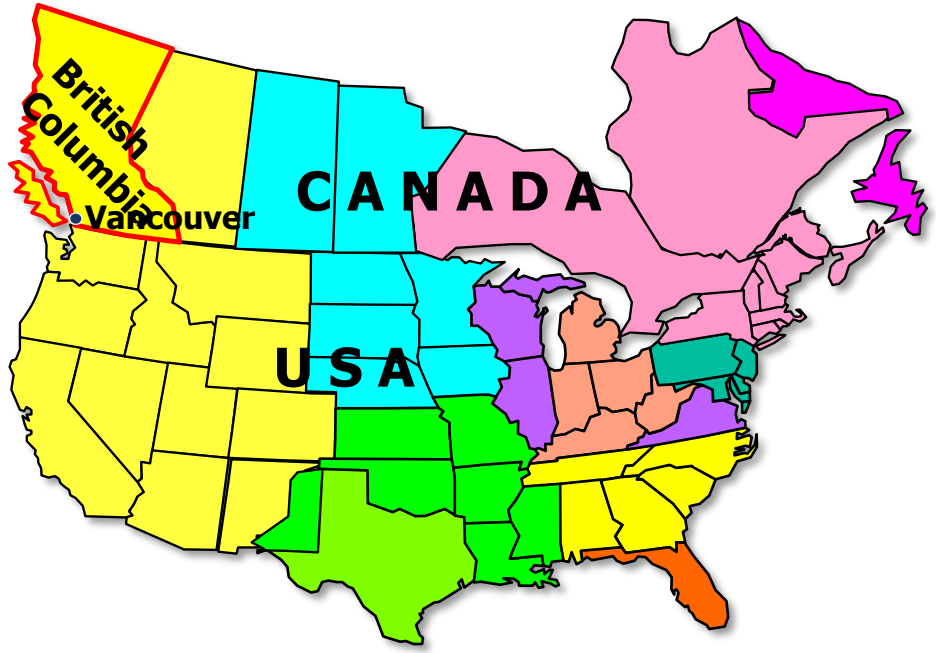
Presented by
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Powertech
The Power of Trust. The Future of Energy.

Powertech WHO ARE WE?

The Power of Trust. The Future of Energy.

- Founded in 1989, and located in the great **Vancouver** area, **BC**, **Canada**
- Subsidiary of British Columbia Hydro and Power Authority
- Large multidisciplinary laboratory, 11-acre site with **34 primary labs**
- Over **40 years** of specialized engineering experience
- **250+** employees, technologists, professional engineers, and PhDs
- Main business includes various types of testing (high voltage/power, hydrogen, and many more), power engineering consulting, power software (DSATools™), and hydrogen
- Management system registered to ISO 27001, ISO 9001, ISO 14001, and ISO 45001 and accredited to ISO/IEC 17025 for specific tests





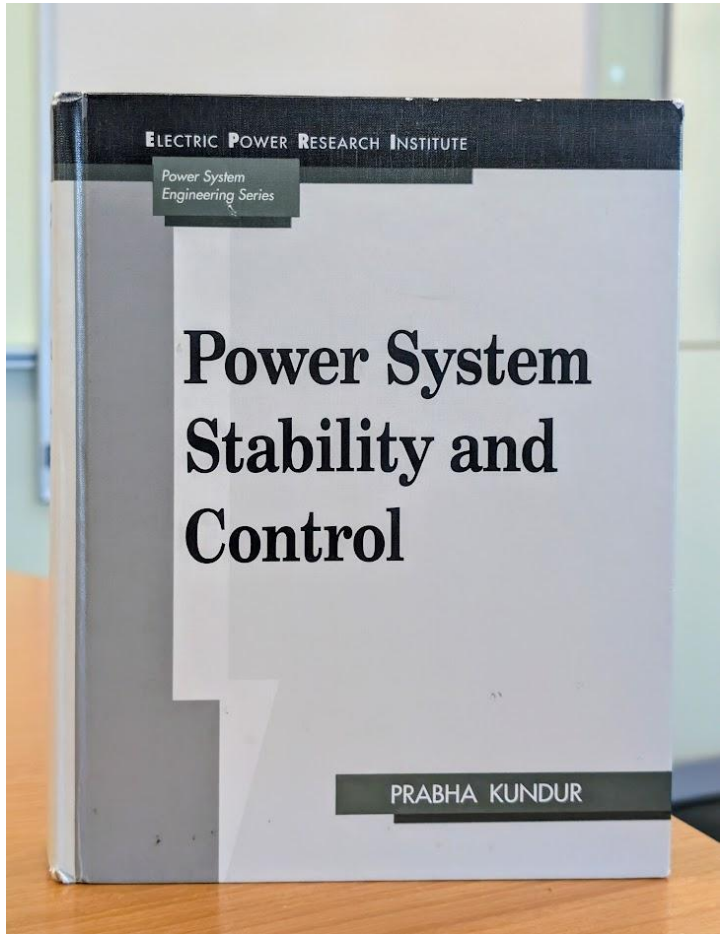
OVERVIEW

- An introduction to DSATools™
- On-line Dynamic Security Assessment (DSA) Overview
- On-line DSA Application Examples

AN INTRODUCTION TO DSATOOLS™

DSATOOLS – BACKGROUND

- Powertech's software team was founded by Prabha Kundur, President and CEO of Powertech from 1994 to 2006
- Powertech started to develop DSATools™, a suite of power system analysis software programs, since mid-1990s
- 25+ years history of **online** and offline applications
- **Version 26** was released July 2026
- Software team provides comprehensive services including software upgrades, software customizations, model development, technical support, and training.



DSATOOLS – CORE MODULES

PSAT – Powerflow and Short-circuit Analysis Tool

VSAT – Voltage Security Assessment Tool

TSAT – Transient Security Assessment Tool

SSAT – Small Signal Analysis Tool

DSA Manager – On-line DSA interface module

DSATOOLS ADD-ON MODULES

UDM Editor – User-defined model editor

DSAOA – Output analysis for VSAT and TSAT

CDT – Control design toolbox for PSS design

SSR – Sub-synchronous resonance analysis

TRI – TSAT-RTDS™ Interface for hybrid simulation

TPI – TSAT-PSCAD™ Interface for hybrid simulation

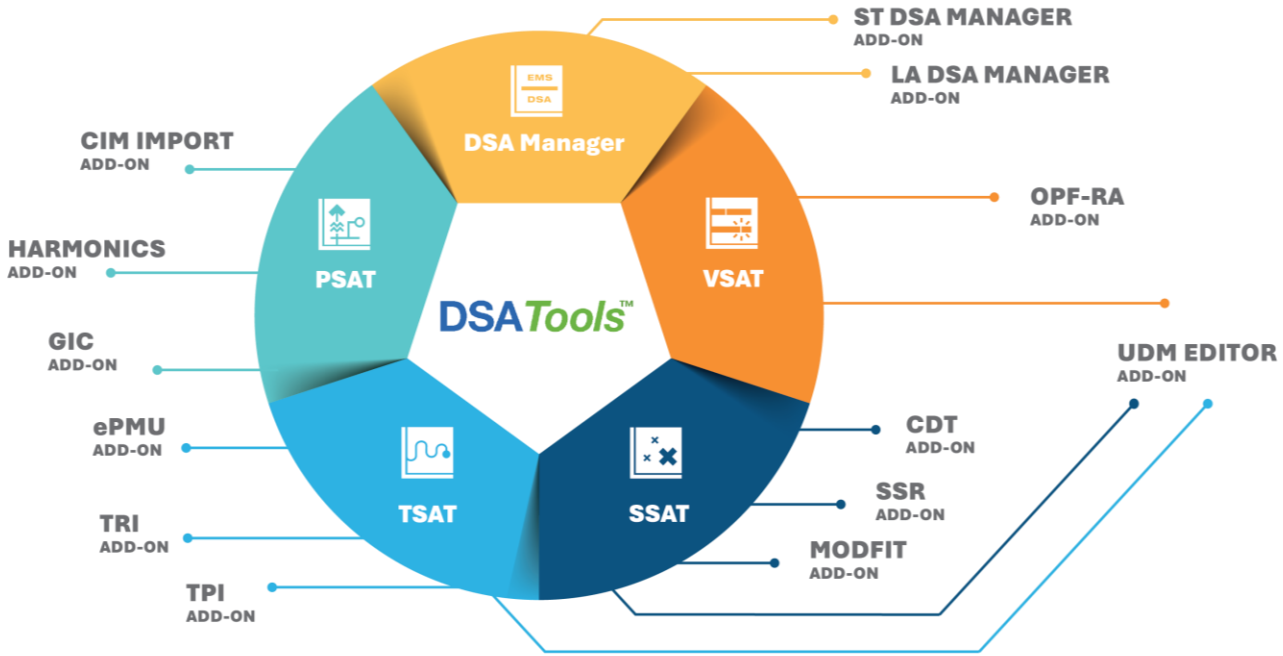
ST DSA Manager – Study mode manager for on-line DSA

LA DSA Manager – Look Ahead manager for on-line DSA

Harmonics – Harmonic analysis

MODFIT - Model fitting tool to incorporate black-box PSCAD models in small signal stability studies

CIM Import – CIM data importer



ePMU – Generator of simulated PMU data (IEEE C37.118)

OPF-RA – Remedial action determination using OPF

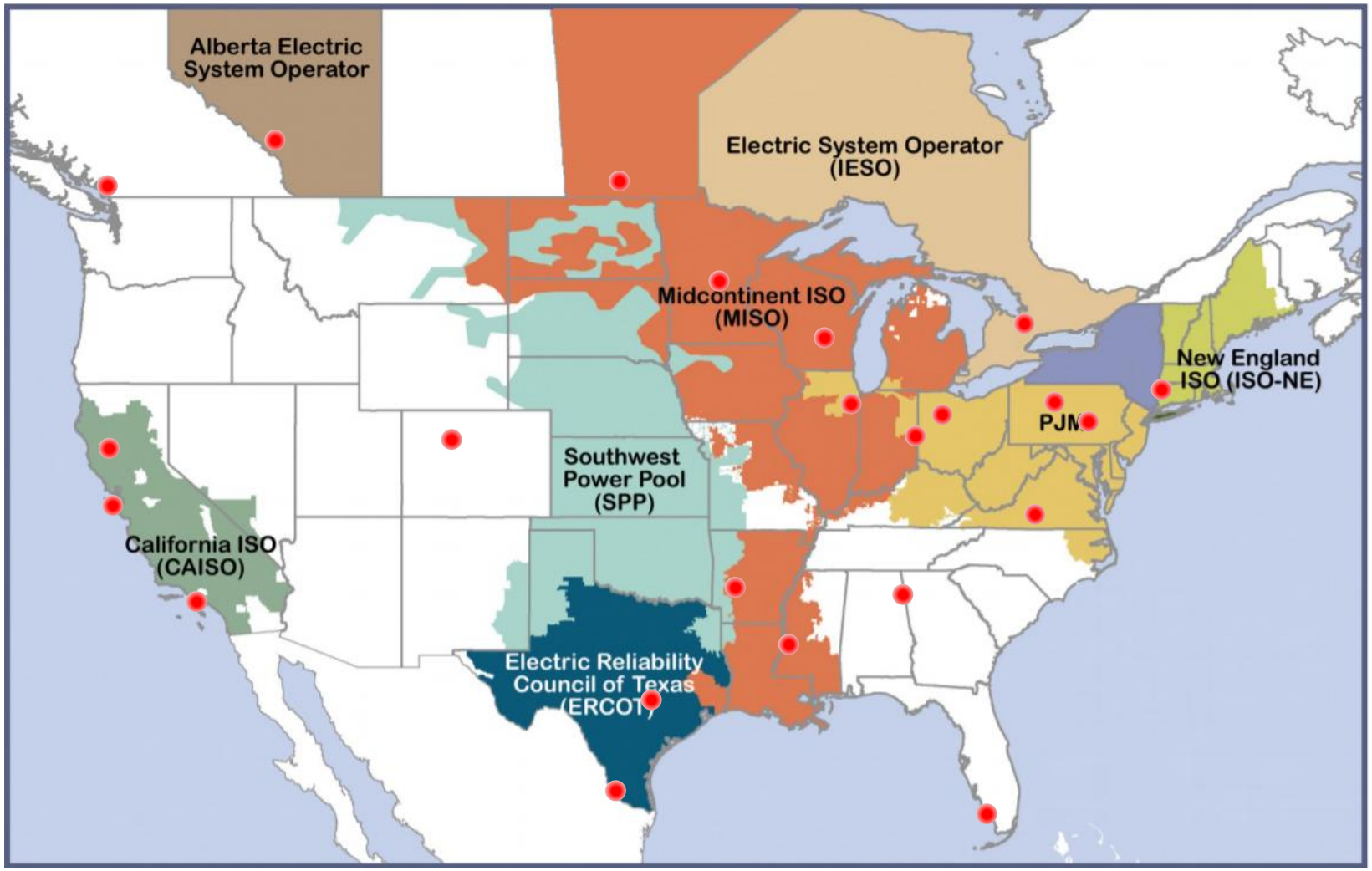
GIC - Geomagnetically Induced Current (GIC) calculation and Geomagnetic Disturbance Event Analysis Tool

WORLDWIDE DSATOOLS™ USERS



ON-LINE DSATOOLS™ APPLICATIONS IN NORTH AMERICA

- On-line DSA installations in North America – including 8 ISOs and major utilities



SELECTED ON-LINE DSATOOLS™ USERS - WORLDWIDE

	Licensed Products	Year of Licensing
AEMO (Australia Energy Market Operator, Australia)	VSAT	2008
KPX (Korea Power Exchange, South Korea)	VSAT/TSAT/SSAT	2012/2016
EirGrid (Ireland)	VSAT/TSAT	2002
EMA (Singapore)	VSAT/TSAT	2012
JEPC (State Grid Jiangsu Electric Power Company, China)	TSAT/VSAT	2018/2019
KEPRI (KEPCO Research Institute, South Korea)	TSAT/VSAT/SSAT	2018
National Grid (Saudi Arabia)	VSAT/TSAT	2014
National Grid ESO (UK)	TSAT/VSAT	2012/2022
NTDC (Pakistan)	VSAT/TSAT	2022
SEB (Sarawak Energy Berhad, Malaysia)	VSAT/TSAT	2012
SONELGAZ (Algeria)	VSAT/TSAT	2022
Statnett (Norway)	VSAT/TSAT	2011
TNB (Tenaga Nasional Berhad, Malaysia)	VSAT/TSAT	2005
Transelectric (Ecuador)	VSAT/TSAT	2015
Transpower (New Zealand)	VSAT/TSAT	2005/2010/2013
XM (Colombia)	VSAT	2010

ON-LINE DYNAMIC SECURITY ASSESSMENT (DSA) OVERVIEW

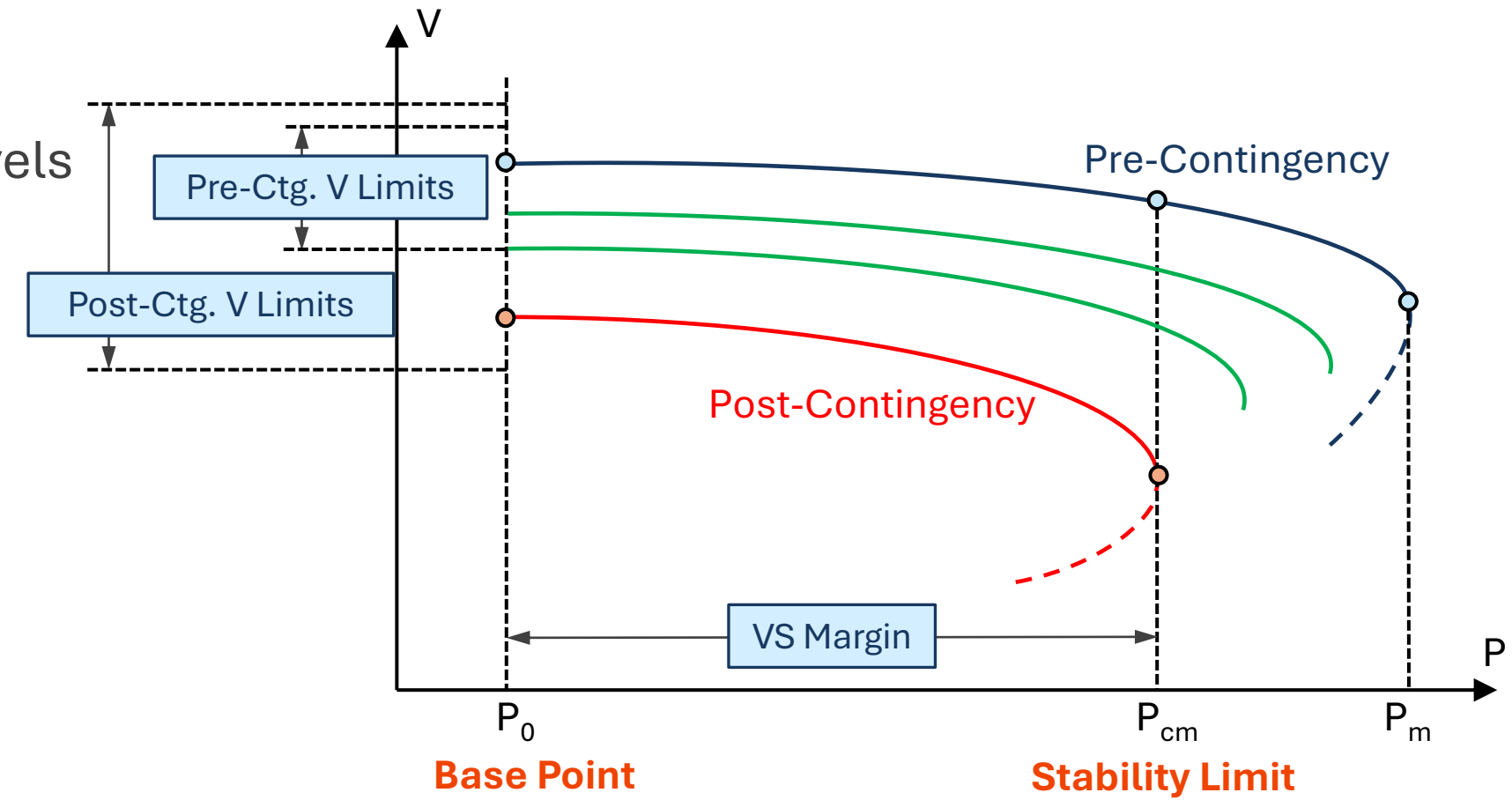
DYNAMIC SECURITY ASSESSMENT

- Required in power system planning and operation
- Insecure conditions may appear as
  - Overloading in transmission equipment
  - Unacceptable voltage deviations
  - Unacceptable frequency deviations
  - Voltage collapse
  - Transient instability
  - Poor or unstable oscillations
  - Frequency instability
  - Sub-synchronous interactions
  - Others

Dynamic Security Assessment (**DSA**) is an evaluation of the ability of a power system to withstand a defined set of contingencies and to survive the transition to an acceptable steady-state condition

SECURITY CRITERIA IMPLEMENTED IN DSATOOLS

- Steady state
  - Thermal loading
  - Steady state voltage levels at pre- and post-contingency conditions
  - VAR reserve
  - Weighted short-circuit ratio, short-circuit ratio
  - Voltage stability



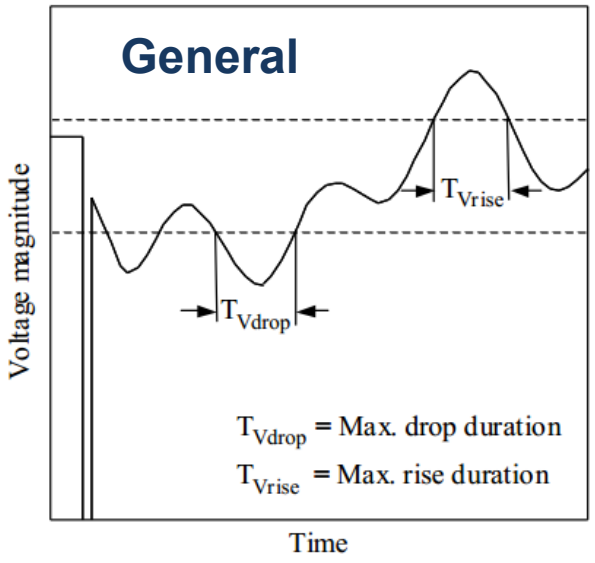
Definition of some steady state voltage security criteria

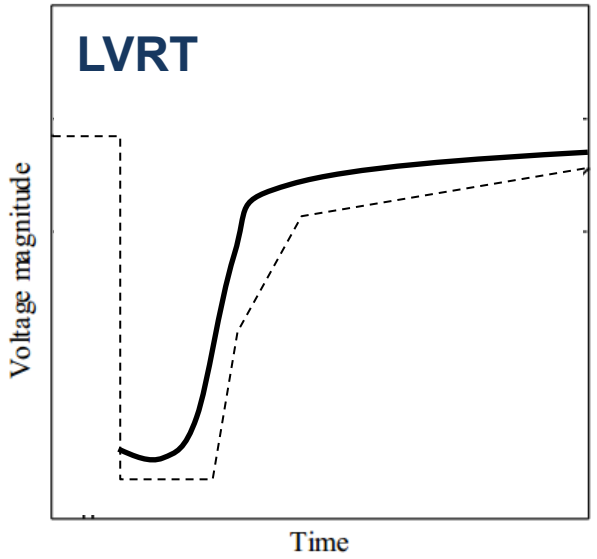
SECURITY CRITERIA IMPLEMENTED IN DSATOOLS (CONT'D)

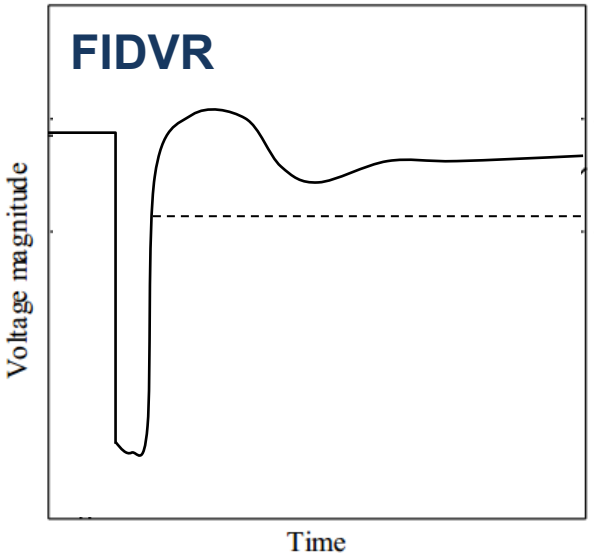
- Dynamic
  - Transient stability: may be based on
    - Power swing, or
    - Power angle
  - MW tripped due to out-of-step
  - Peak-to-peak angle separation
  - Torque on generator shaft
  - Critical clearance time (CCT) of faults
  - Damping factor of critical oscillatory modes
  - Relay margin
  - *…more on next page*

SECURITY CRITERIA IMPLEMENTED IN DSATOOLS (CONT'D)

- Dynamic
  - Transient voltage performance
    - ❖ General voltage dip/rise
    - ❖ LVRT
    - ❖ FIDVR
  - Frequency performance
    - ❖ General frequency dip/rise
    - ❖ Rate of change of frequency (ROCOF)







SECURITY CRITERIA IMPLEMENTED IN DSATOOLS (CONT'D)

- Customizable for different study scenarios
- Can be applied to specific regions of a system
- Can be applied to specific contingencies
- Can be applied to determine transfer limits
- Fully support widely used study guides
- **No lengthy post processing required – TSAT/VSAT/SSAT output detailed violation reports at the end of the computation**
- **Cascading analysis**

WHAT IS ON-LINE DSA?

- **Objective:** to ensure secure operation of a power system
- What's the **difference** from the conventional (off-line) approach?
  - Starts from a **real-time captured** (or forecasted) system condition
  - Considers all applicable security criteria and credible contingencies ("**compliancy**")
  - Include high fidelity system models ("**model-based analysis")**
  - Perform analyses **automatically**, **periodically**, and **efficiently**
  - Present results in simple, straightforward way – security margin, operational limits, remedial actions required, etc.

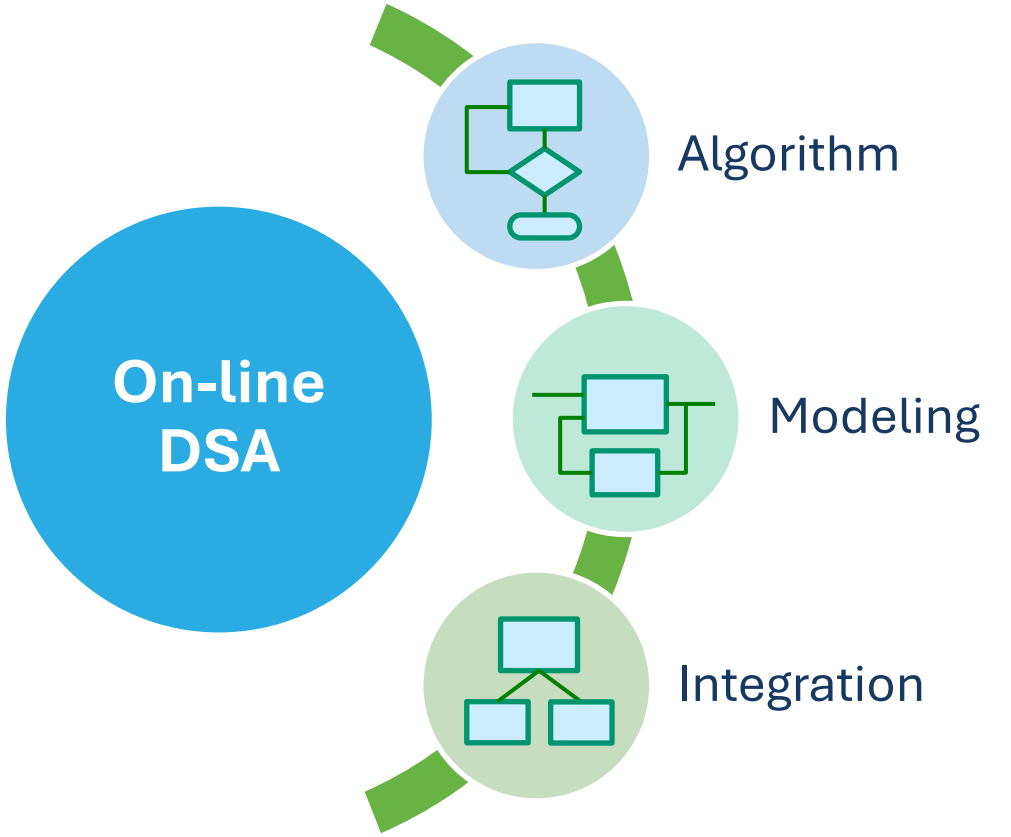
Powertech's on-line DSA software is **proven choice for online DSA** running in many control rooms around the world

WHY DO DSA ON-LINE?

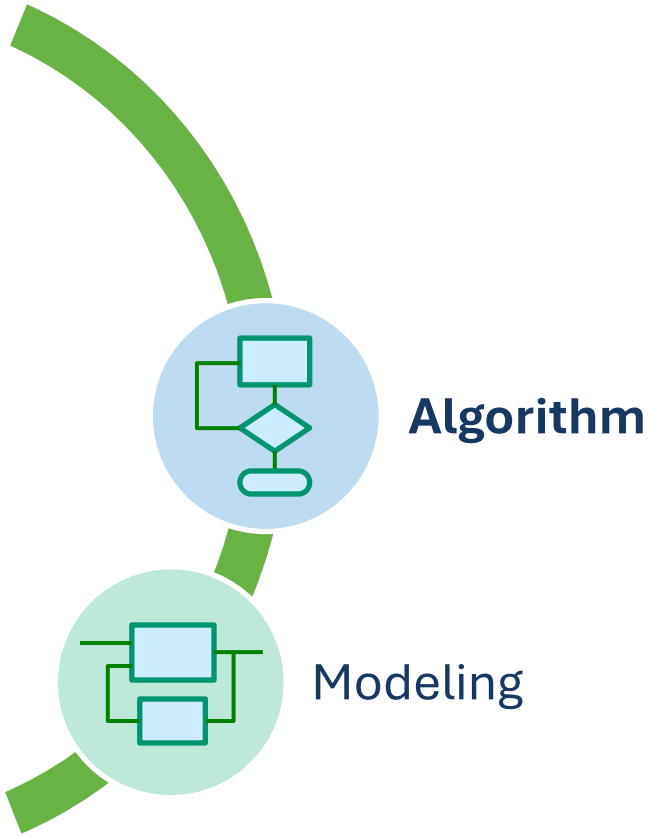
- Improved grid security . . . **prevent blackout!**
- **Better situational awareness** in control rooms
- Dealing with unpredictability and uncertainty in grid operation, especially with **high renewable penetration**
- Maximized use of **transmission capacities**
- Dealing with congestion in power market
- Minimized generation and reserve cost
- Reliability compliance
- Fast and accurate controls
- *More . . .*



THREE KEY ELEMENTS IN ON-LINE DSA



ALGORITHM



Static analysis

- **P-V**
- **Q-V**
- **Continuation powerflow**
- **Modal analysis**
- **Short circuit analysis**
- **OPF**

Dynamic analysis

- **Time-domain simulations**
- Direct methods (EEAC, BCU, etc.)
- **Eigenvalue analysis**
- Trajectory sensitivities
- Sub-synchronous interaction analysis

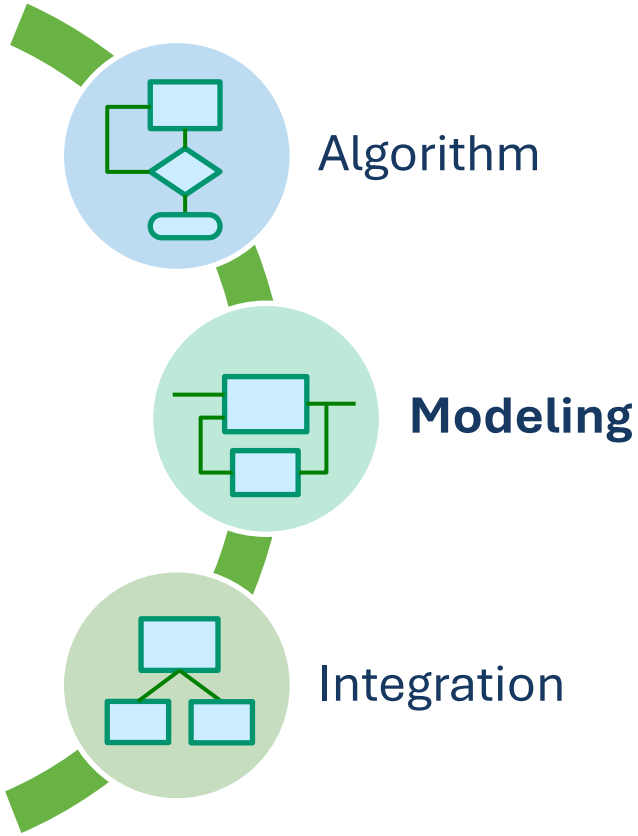
Special analysis

- **Transfer limit analysis**
- **Remedial controls**
- **Distributed/parallel/** cloud computations
- **Real-time closed loop controls**
- Big data analysis
- Probabilistic analysis
- AI methods

Blue – Implemented in DSATools™

New methods are being proposed and applied continuously

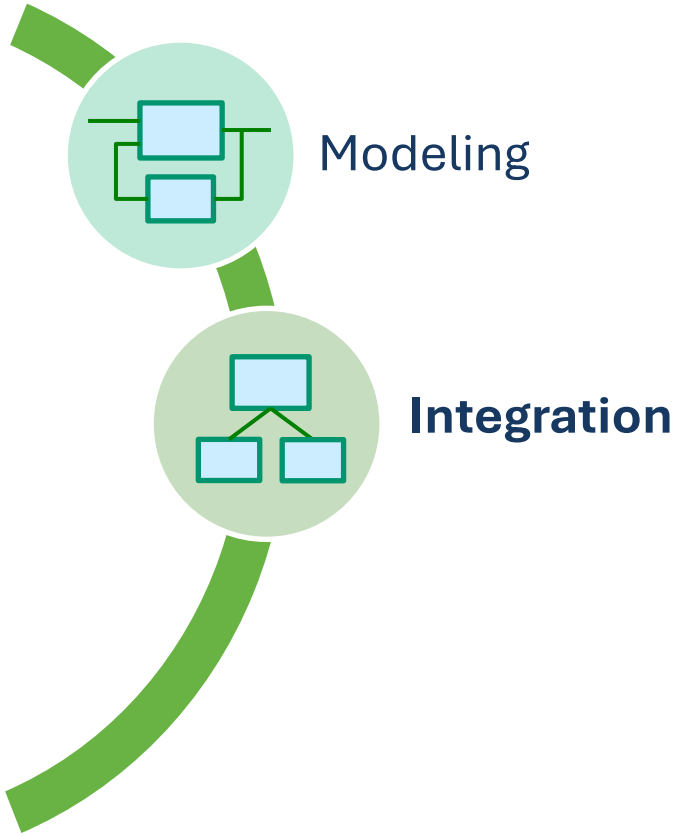
MODELING



In addition to the models considered in the conventional off-line system planning studies, additional models or modeling techniques are required for on-line DSA

- **Examples:**
  - Models need to handle various operating points
  - Node/breaker modeling
  - RAS and SPS models
  - Realistic contingency sequences
  - Real-time equipment status
  - Dynamic rating
  - Real-time renewable status
  - External system equivalencing

INTEGRATION



Integrated Control Room Application

- Real-time
- Forecast
- Look-ahead
- Market
- PMU/WAMS
- Training
- Testing
- Study
- Redundant/Backup
- Result depository

On-line DSA

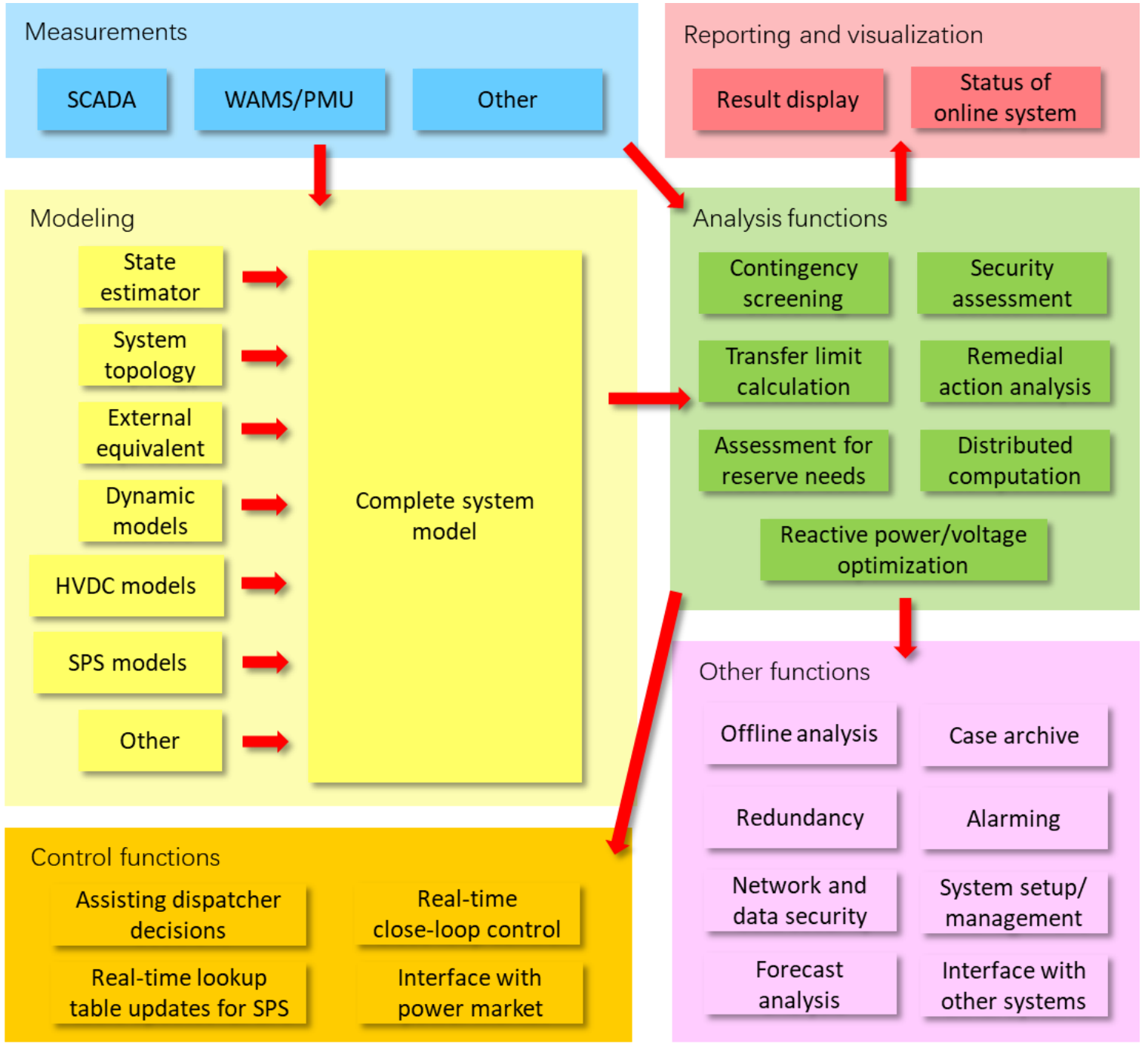
- RT model recondition/ match
- Auxiliary data management
- Resource and data sharing
- Analysis scenario creation
- Performance monitoring
- Result processing and display
- Failover and security

Auxiliary Databases

Computation Engines

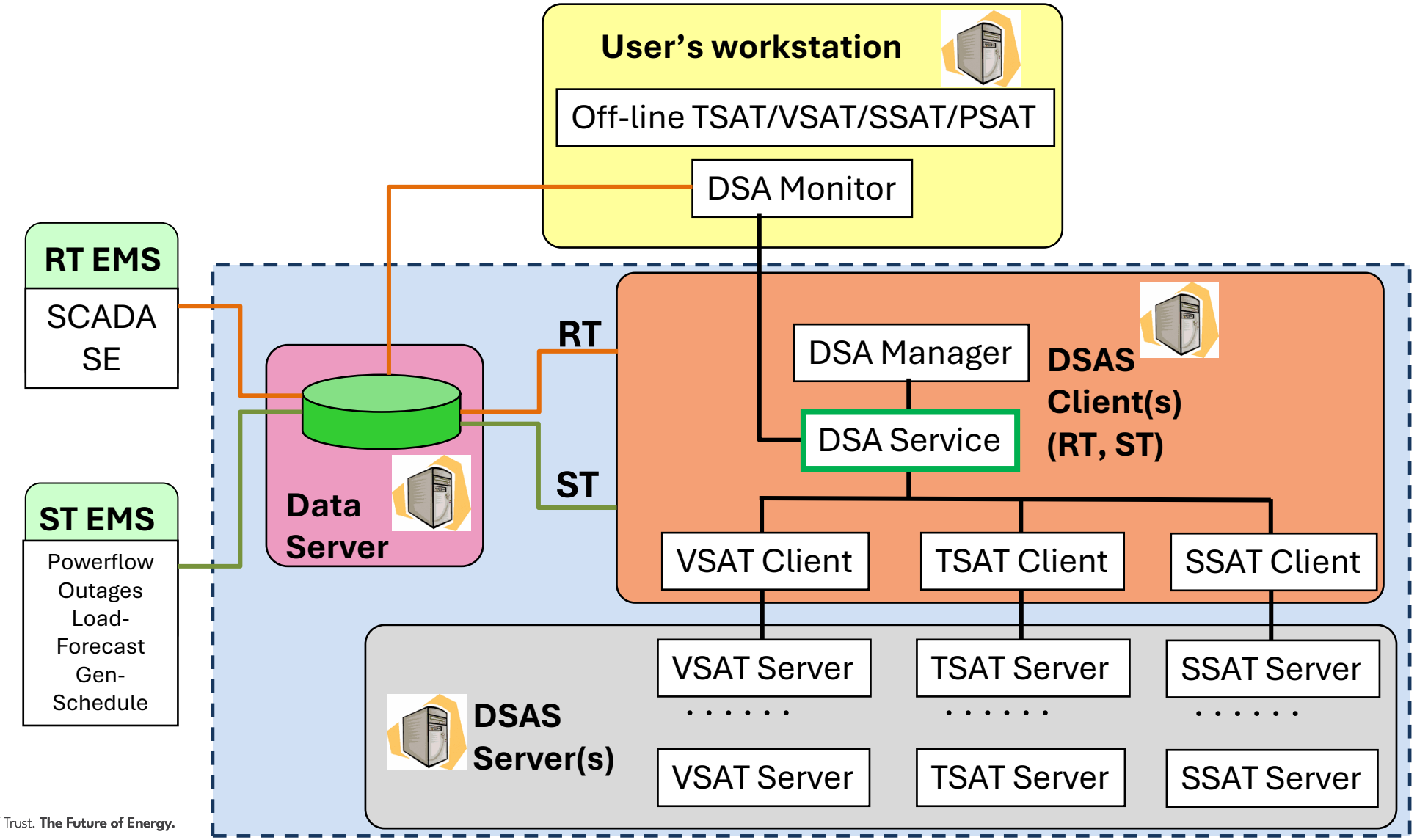
- Distributed/parallel / cloud computing

ON-LINE DSA FUNCTIONAL MODULES



DSATools™ Dynamic Security Assessment Software

ON-LINE DSA - SOFTWARE / HARDWARE OVERVIEW



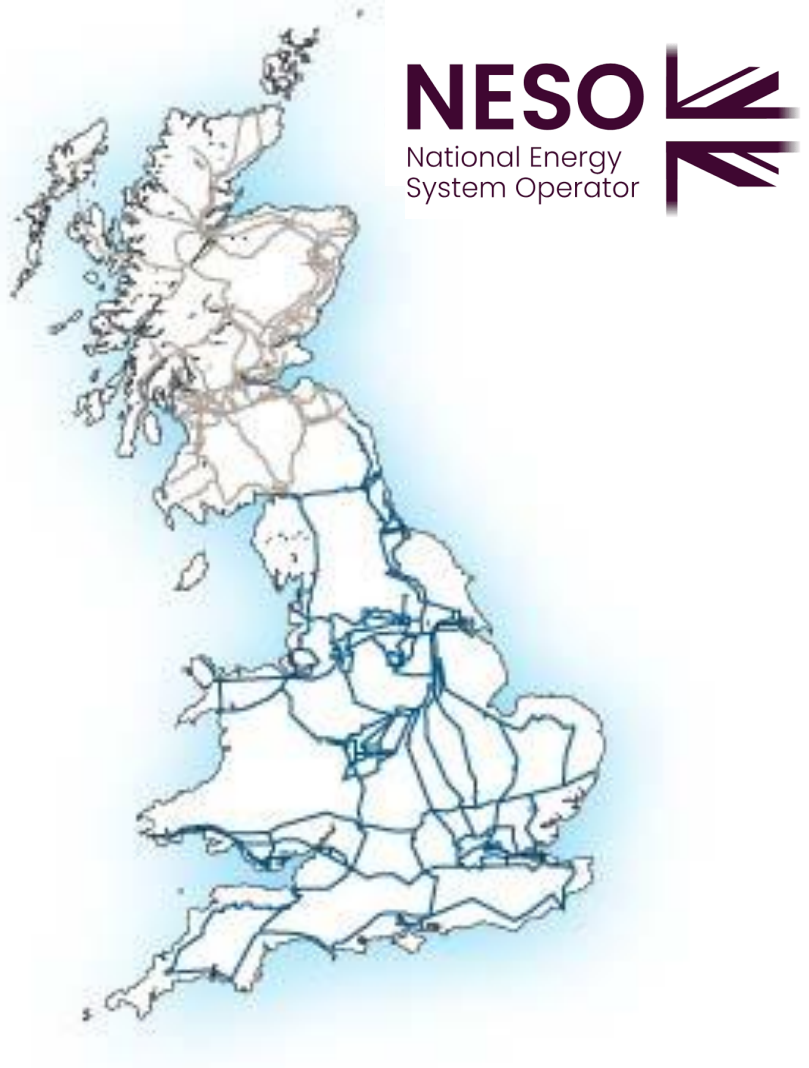
ON-LINE DSA APPLICATION EXAMPLES

EXAMPLE USE CASES & APPLICATIONS

- System security status and margin monitoring
- Power transfer limit determination
- Impact assessment of renewables on system security
- Verification of system protection schemes (SPS)
- System inertia estimation
- System strength assessment
- IBR stability assessment
- Recommendations for preventive/corrective control actions
- Transaction settlement in power market
- Determination of active and reactive power reserve
- Scheduling of equipment maintenance
- Support to PMU/WAMS applications
- Calibration and validation of power system models
- Preparation of models for system studies
- Post-mortem analysis of incidents
- System restoration

SYSTEM SECURITY STATUS MONITORING

- Motivation for the NESO (UK) on-line stability analysis
  - 60 GW island system with DC Interconnections to Europe
  - Renewables brought more uncertainty
  - **Off-line studies no longer adequate**
- Capabilities of the NESO on-line DSA
  - Including both transient stability and damping criteria
  - 3,000 bus system model
  - Simulate **2,000 contingencies** (N-1, N-2, detailed), each contingency simulated for 20 seconds
  - **15-min** computation cycle with **200 CPU cores**

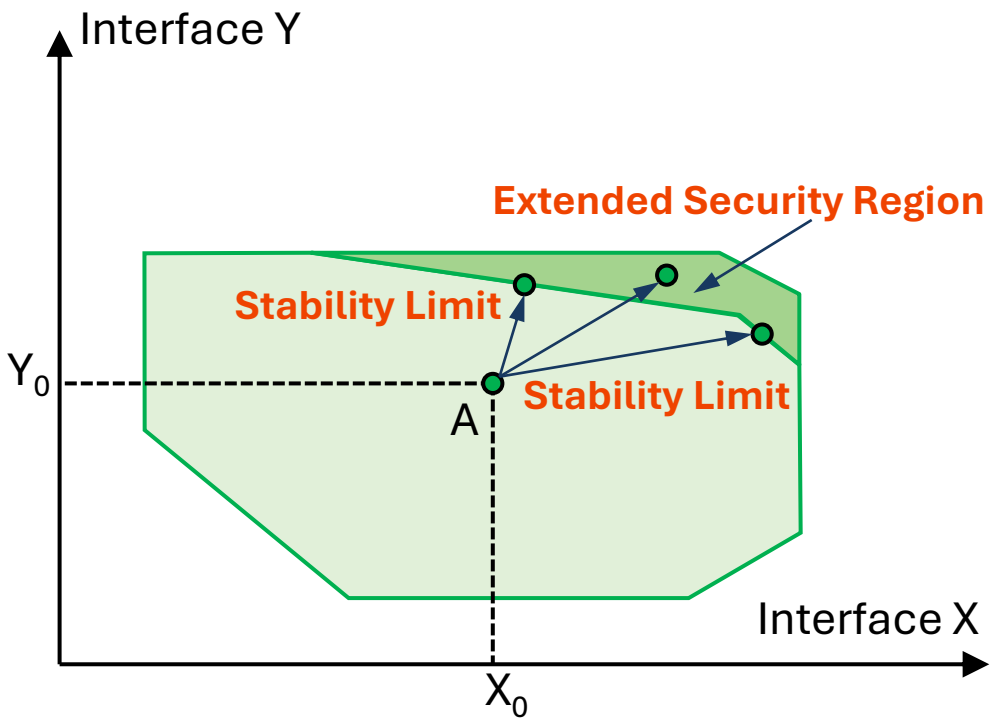


SYSTEM SECURITY STATUS MONITORING (CONT'D)

- A special **study mode** enables engineers to quickly perform studies for various scenarios based on real-time system conditions
- Benefits achieved
  - Transient stability risks identified, which were not previously known
  - Damping risks identified including one involving a group of 8 GW generation, which were not previously known
  - Instability risks identified at special operation conditions (hydro units in pumping mode, circuit switching for voltage control, etc.)

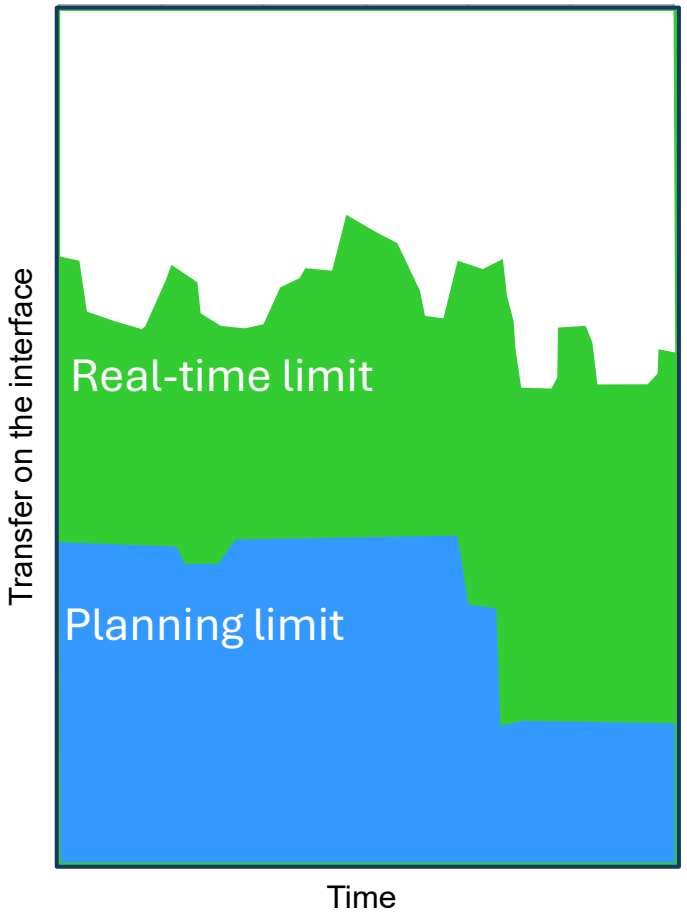
POWER TRANSFER LIMIT ANALYSIS (SECURITY LIMIT)

- Assume that a system is operating at a condition (“A”)
  - Measured by some system parameters (e.g. interface flows)
  - Determined to be secure for all credible contingencies
- We need to know what is the secure region within which the system can move
  - Or which direction to move to achieve maximum transfer (e.g. export/import)
- If the system needs to move to an insecure point for some reason, what needs to be done to ensure the system security
  - Determined, for example, by adjusting special protection system settings



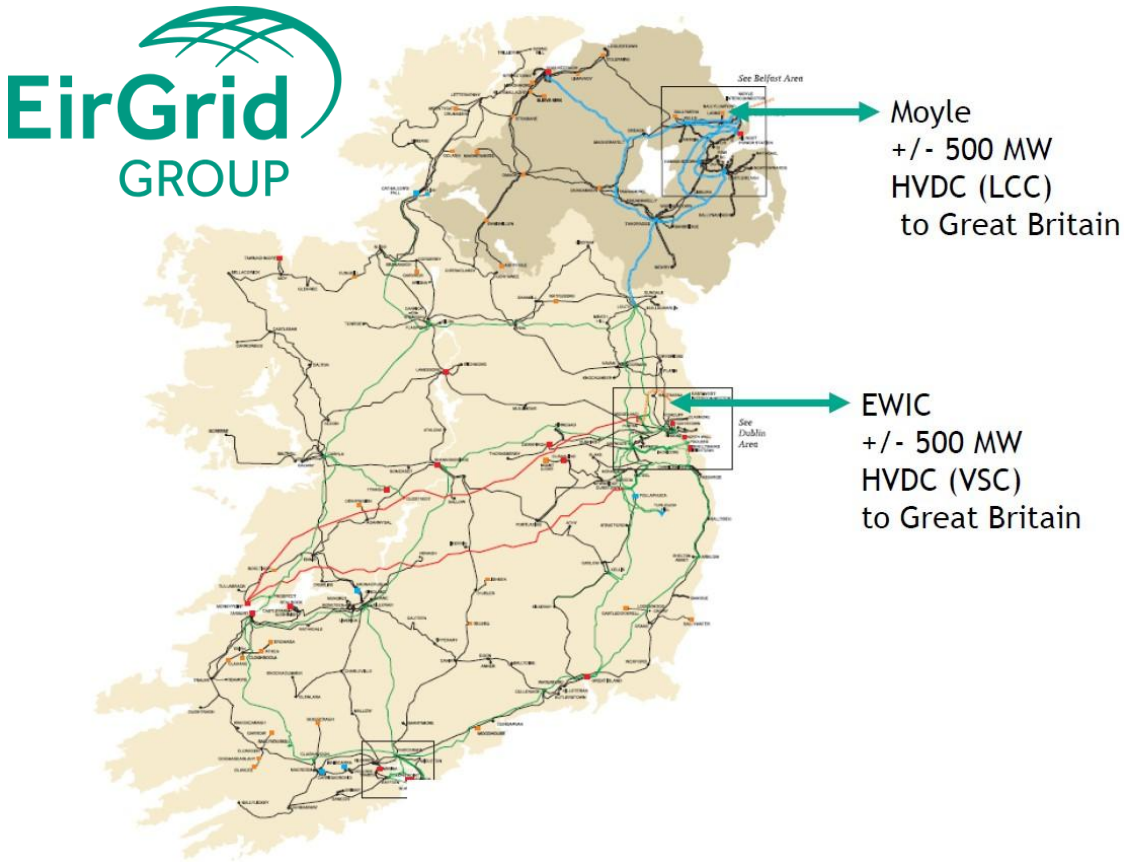
BENEFITS OF TRANSFER LIMIT ANALYSIS

- Increased degree of system security
- More opportunities for power market participants
  - Real-time calculated limits are generally higher than the planning limits
- Lower cost for congestion management
  - On-line DSA helped Midcontinent ISO reduce the annual congestion cost of a key stability-limited interface from $31 million in 2014 to just $0.1 million in 2016.
- Higher level of renewable export
  - A study by ERCOT (the ISO in Texas, USA) discovered that the limits based on-line DSA could allow as much as 400 MW more wind export from its western region (more than 10% of the total transfer capacity)



RENEWABLE INTEGRATION

- **EirGrid** is the Irish national grid company
- Small system but with high wind power penetration
- The question: what is the highest amount of wind generation allowed at any given instant of time?
- Subject to thermal, voltage, frequency, and stability criteria





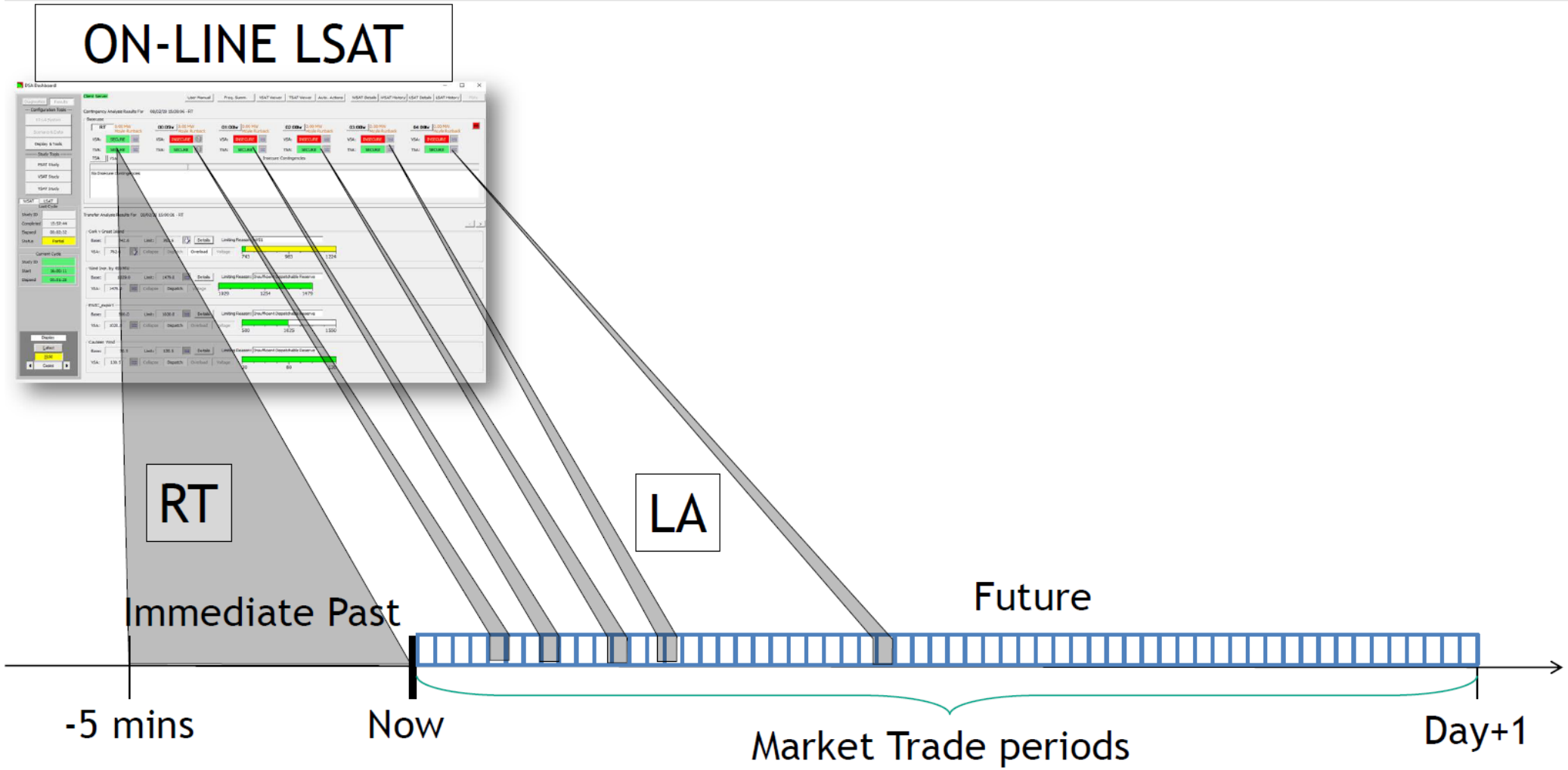
RENEWABLE INTEGRATION (CONT'D)

- Wind Security Assessment Tool (WSAT) was developed in 2010 at EirGrid control center as a real-time application
  - Based on Powertech's VSAT and TSAT
  - Provides max allowable wind generation for Real-time condition
- Look-Ahead DSA (LSAT) was developed in 2020, to replace the WSAT system
  - Combines both Real-time (RT) and Look-Ahead (LA) scenarios.
- Why LSAT?
  - Operational complexity increases with level of non-synchronous renewables
  - Forward looking DSA capabilities were identified as a requirement to increase SNSP above 65%

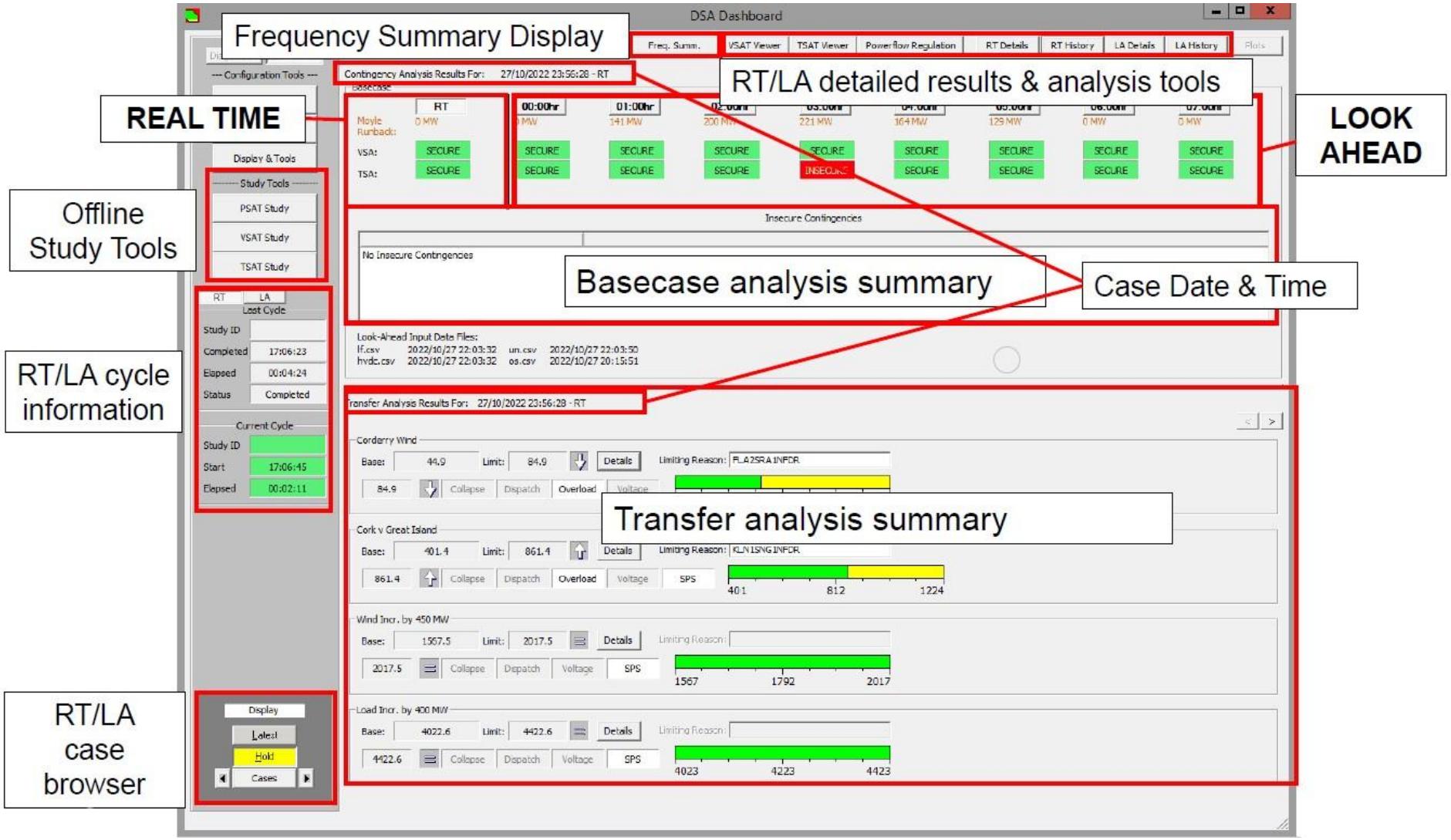
System Non-Synchronous Penetration

$$SNSP = \frac{\text{Wind + Solar + HVDC Imports}}{\text{Demand + HVDC Exports}}$$

RENEWABLE INTEGRATION (CONT'D)

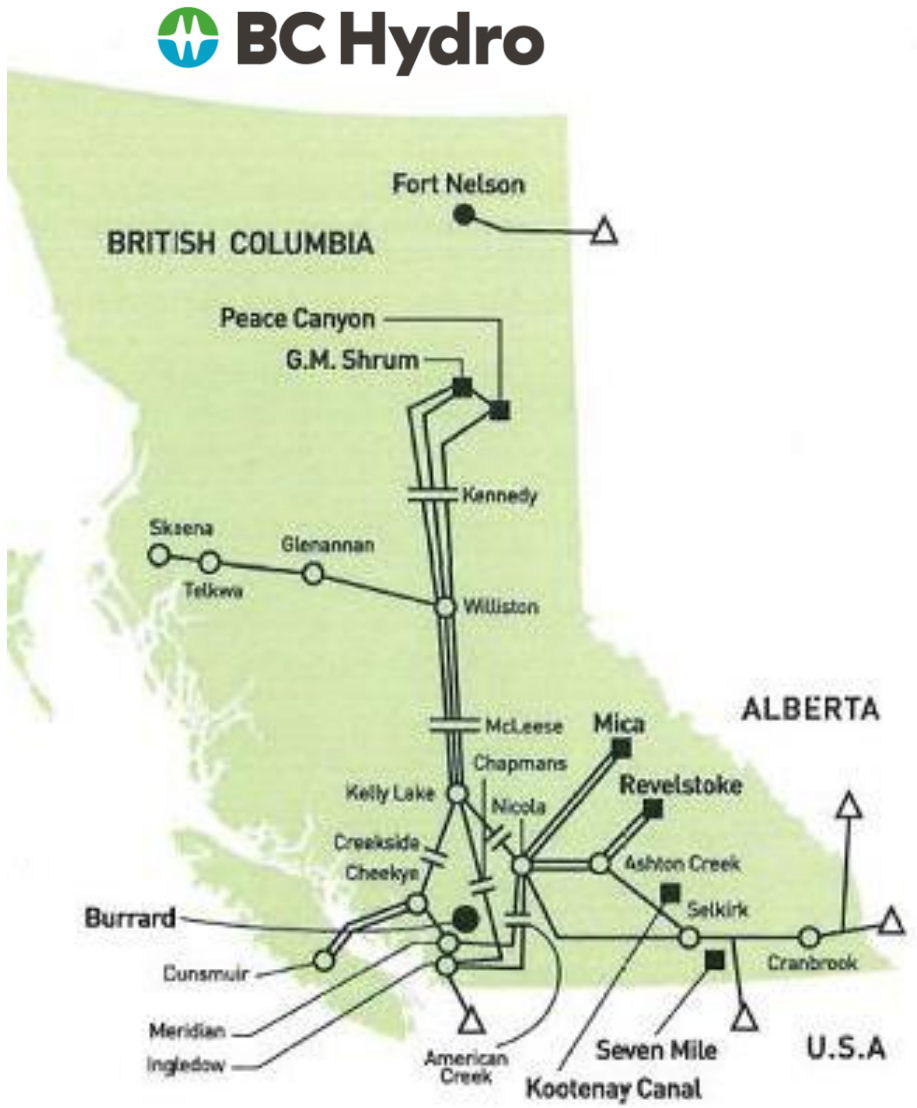


RENEWABLE INTEGRATION (CONT'D)



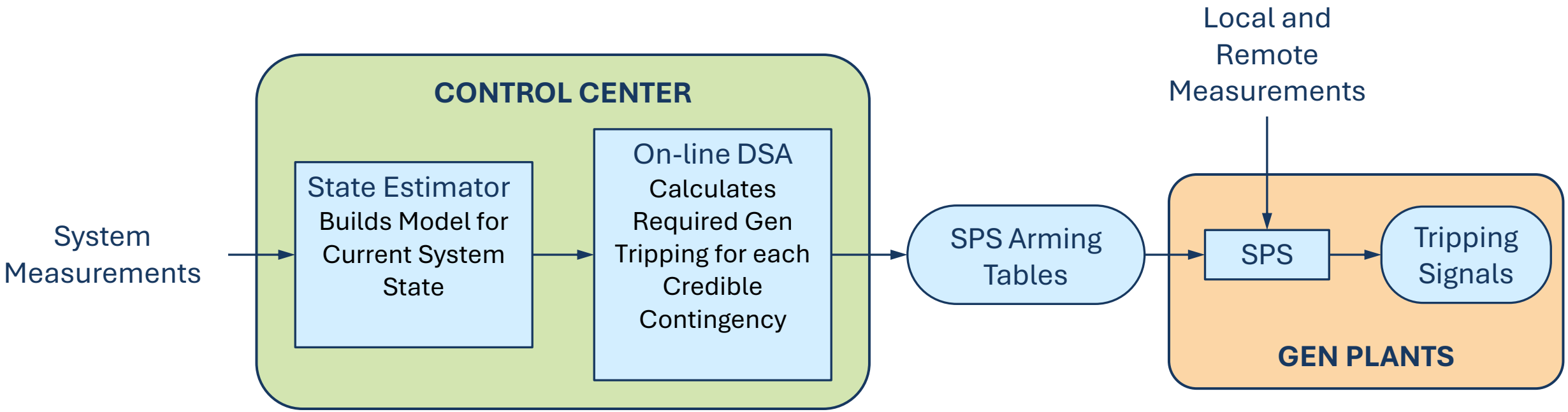
REAL-TIME SPS ARMING

- **BC Hydro**'s major transfers are constrained by stability
- Special Protection Systems (SPS, mainly generator tripping) are used to ensure system security
- **Real-time generator tripping arming** requires on-line lookup table
- A power flow pattern matching (PM) method has been used which relies on off-line computed lookup tables
- This no longer meets the demand from system operation
- The PM method is being phased out with the on-line DSA technology
- Voltage, frequency, and transient criteria are examined



REAL-TIME SPS ARMING (CONT'D)

- The on-line DSA system runs every 4 minutes to determine
  - Gen tripping required for the current transfer
  - Gen tripping required to achieve max transfer



REAL-TIME SPS ARMING (CONT'D)

CONTINGENCY	SET	TRANSFER TOTAL	REQUIRED SHEDDING	ARMED SHEDDING	[redacted] 1	2	3	4	5	6	7	8
[redacted]		277.2	142.2	142.5								
[redacted] 1P		277.2	0.0	0.0								
[redacted]		318.5	183.5	187.8								
[redacted] 1P		318.5	22.3	82.2								
[redacted]		347.3	212.3	212.3								
[redacted] 1P		347.3	31.4	82.2								
[redacted]		-176.4	416.2	546.3								
[redacted]		-172.7	416.2	546.3								

BC Hydro

OLTSA Update Time: 05-Oct-2026 14:52:21 OLTSA Svr found

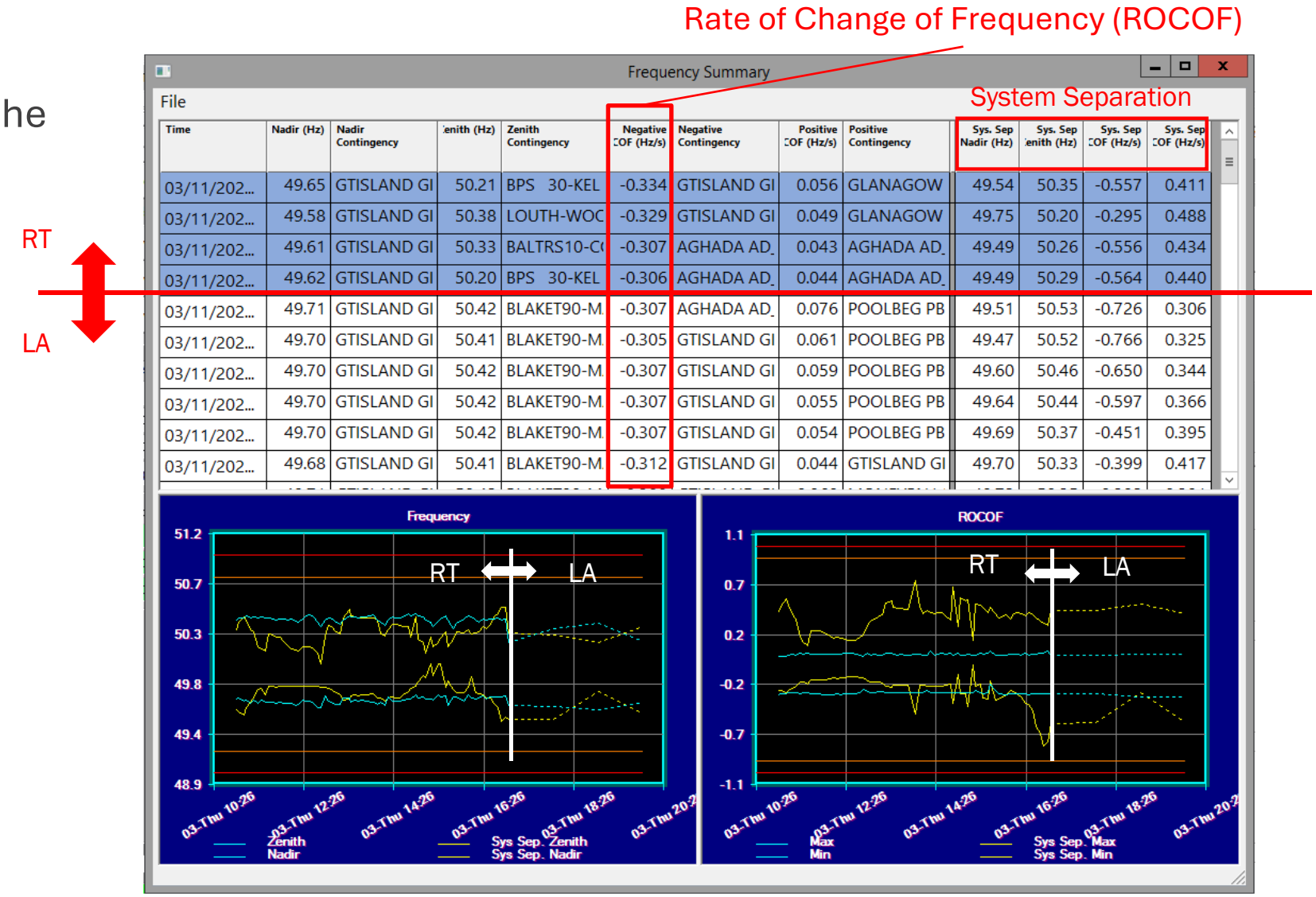
Scenario ID / Ctg.	Current Armed	TSAPM	OLTSA	Required Genshed OLTSA	Armed Genshed OLTSA	Required Genshed TSAPM	Armed Genshed TSAPM
PEACE	05-Oct-2026 14:00:16						
[redacted]	Insecure LR Rise	Insecure LR Rise	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	362.9	525.5
[redacted]	Insecure LR Marg LR Drop LR Rise LR Frq LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	1113.1	1265.4
[redacted]	Insecure LR Marg LR Drop LR Rise LR Frq LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	1113.1	1265.4
[redacted]	Insecure LR Rise	Insecure LR Rise	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	362.9	525.5
[redacted]	Insecure LR Marg LR Drop LR Rise LR Frq LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	1113.1	1265.4
[redacted]	Insecure LR Rise	Insecure LR Rise	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	362.9	525.5
[redacted]	Insecure LR Rise	Insecure LR Rise	Secure	0.0	0.0	0.0	0.0
[redacted]	Insecure LR Marg LR Drop LR Rise LR Frq LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	Insecure LR Rise	1503.0	1514.3	0.0	0.0
[redacted]_7	Insecure LR Marg LR Drop LR Rise LR Frq LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	Insecure LR Marg LR Drop LR Rise LR Frq	1503.0	1514.3	0.0	0.0
[redacted]	Insecure LR Rise	Insecure LR Rise	Secure	448.0	579.9	1153.4	1336.2

SYSTEM INERTIA ESTIMATION

- With more and more renewable energy resources integrated, **the System Inertial Response** becomes a major concern of system operators
- Three approaches have been used to estimate the system inertia
  - Inertial estimation based on PMU measurement
    - Not depending on models
    - Can not deal with what-if or forecast conditions
    - Not suitable for estimating inertia response under large disturbances
  - Calculate the weighted sum of inertia of generators and motors
    - Synthetic inertia from BESS etc. is ignored
  - Estimate the system inertia by checking the system frequency responses from **time domain simulation**
    - **EirGrid, Transpower NZ** and **California ISO** are using this approach in their On-line DSA

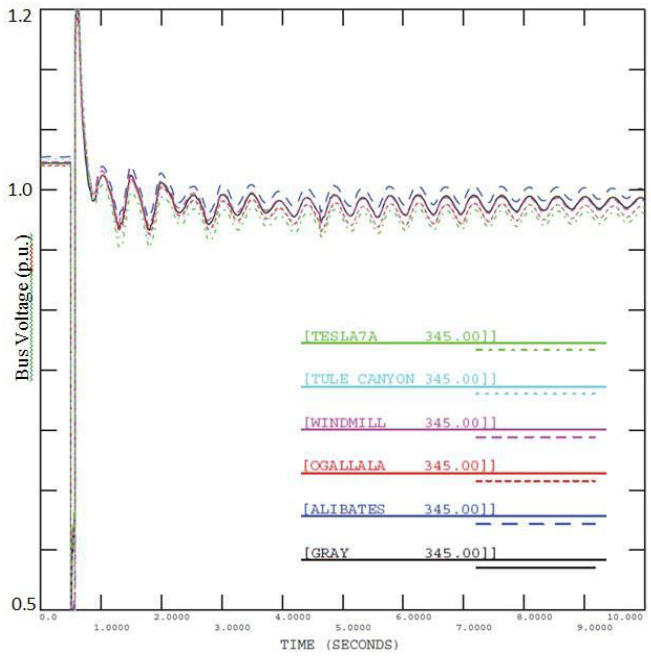
SYSTEM INERTIA ESTIMATION (CONT'D)

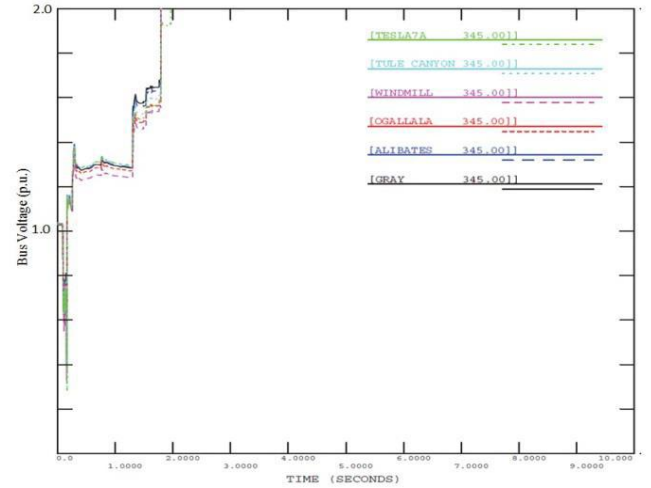
- Frequency Summary Table of the EirGrid LSAT



SYSTEM STRENGTH MONITORING

- It is challenging to integrate Inverter-Based Resource (IBR) to a weak AC system
- Disturbances occurring in a weak system can cause oscillations and cascading over voltages
- System strength reflects system voltage sensitivity to disturbances
- System strength can be measured by short circuit ratio (**SCR**), weighted short circuit ratio (**WSCR**), or effective short circuit ratio (**ESCR**)
- The SCR shows how strong a bus is with respect to the resource capacity

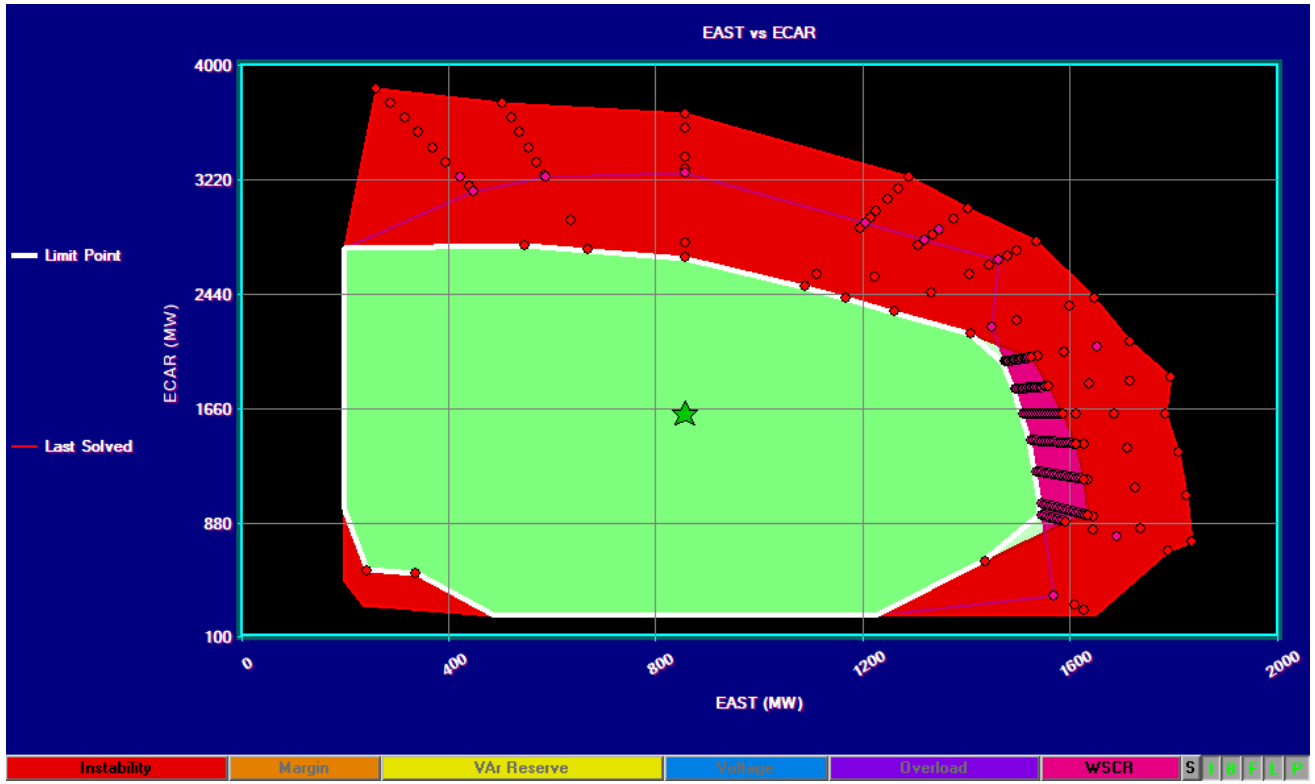




SYSTEM STRENGTH MONITORING (CONT'D)

- DSATools™ includes built-in **WSCR** and **SCR** assessment criteria.
- System strength can be evaluated under both real-time operating conditions and forecasted scenarios, including planned outage or contingency conditions.
- ERCOT uses this capability to continuously monitor several wind-rich regions on a 24/7 basis, where wind generation output is constrained by available system strength.

2-D transfer limit diagram showing the power transfer is limited by low system strength and voltage collapse



IBR STABILITY ASSESSMENT

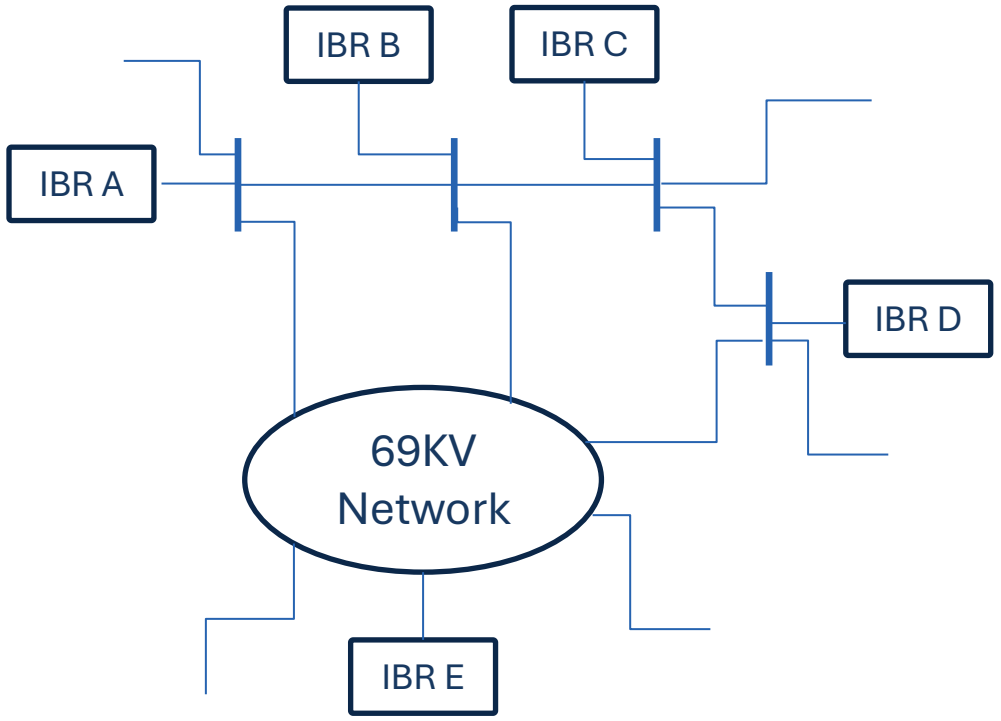
- DSATools™ **SSAT** module is a powerful tool for studying electromechanical oscillations between 0.1Hz to 5Hz

- Increasing prevalence of **IBR-driven instability**
  - Sub-synchronous oscillation frequencies from **3-30Hz** (and above)

- **SSR** mode in SSAT captures the high frequency dynamic behavior, allowing accurate analysis of oscillation modes at sub-synchronous frequencies:
  - **Sub-synchronous Control Interactions (SSCI)**
  - **Weak Grid Instability**
  - Sub-synchronous Torsional Interactions (SSTI)
  - Induction Generator Effect

IBR STABILITY ASSESSMENT (CONT'D)

- **Weak-grid pocket** containing **5 black-box PSCAD models**
- **ModFit** module used to convert the PSCAD black-box models into **SSAT-compatible models**
- Study area size:
  - PSCAD case: **240 buses**
  - SSAT/SSR case: **713 buses**
- Study covering **9 contingencies** across **6 dispatch scenarios**



IBR STABILITY ASSESSMENT (CONT'D)

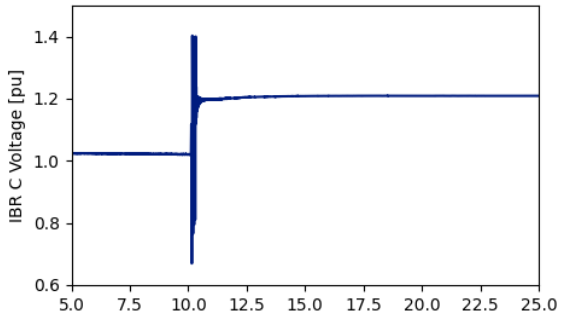
IBR A and IBR B Out

Contingency 3

SSAT/SSR:

Frequency	Damping
26.26 Hz	-18.85 %

PSCAD:



Brief 20 Hz oscillation, then IBR C trips

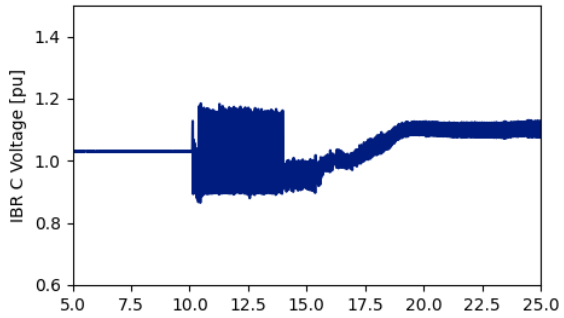
IBR A and IBR B Out

Contingency 5

SSAT/SSR:

Frequency	Damping
33.80 Hz	-7.81 %

PSCAD:



Sustained 30 Hz oscillation

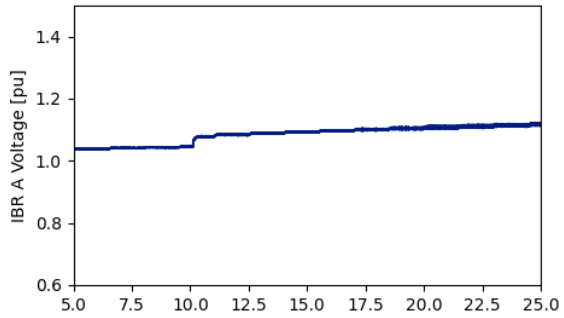
IBR B and IBR C Out

Contingency 5

SSAT/SSR:

Frequency	Damping
43.67 Hz	14.21 %

PSCAD:



Computation Time

SSAT/SSR run (per ctg): **1.6s**

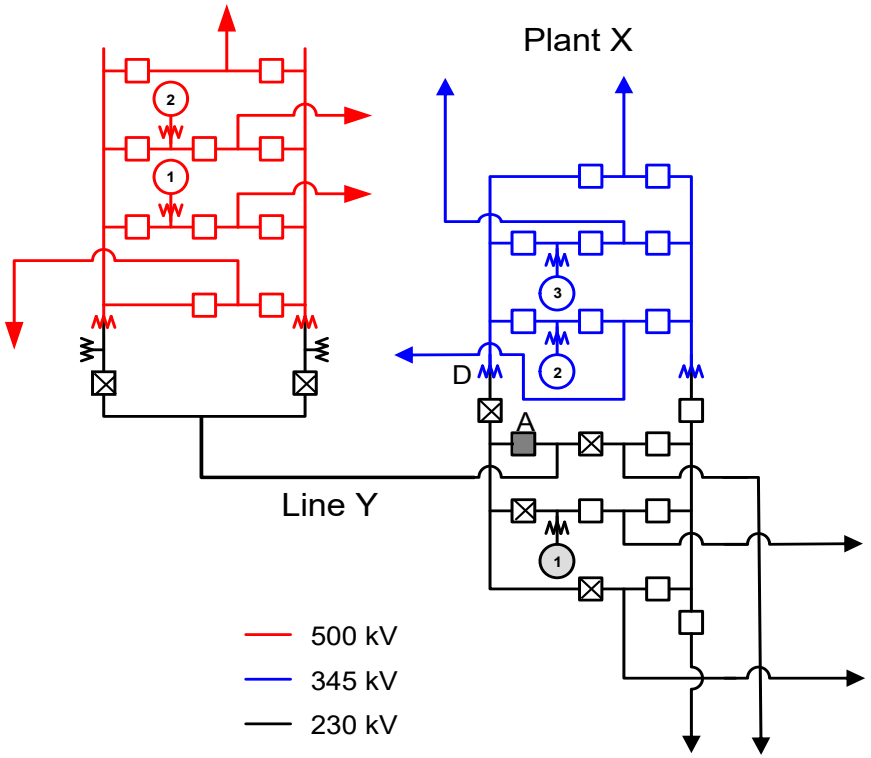
Total (6x9 ctgs): **1.44 m**

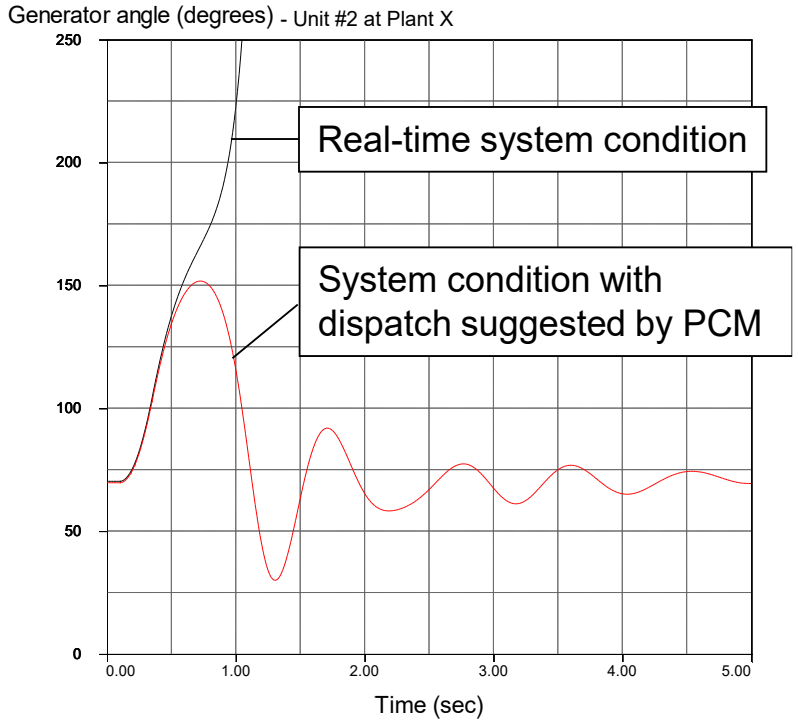
PSCAD run (per ctg): **3h38m**

Total (6x9 ctgs): **196h36m**

RECOMMEND REMEDIAL CONTROL ACTIONS

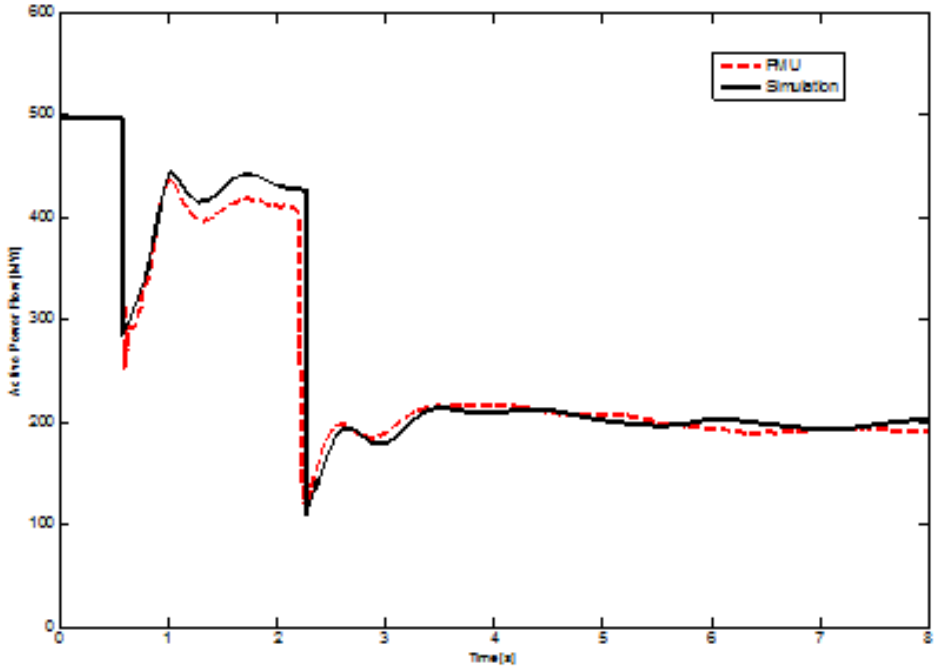
- A case obtained from an actual on-line DSA system which includes a Preventive Control Measure (PCM) module
- The PCM module was activated to investigate a transiently unstable contingency





MODEL VALIDATION

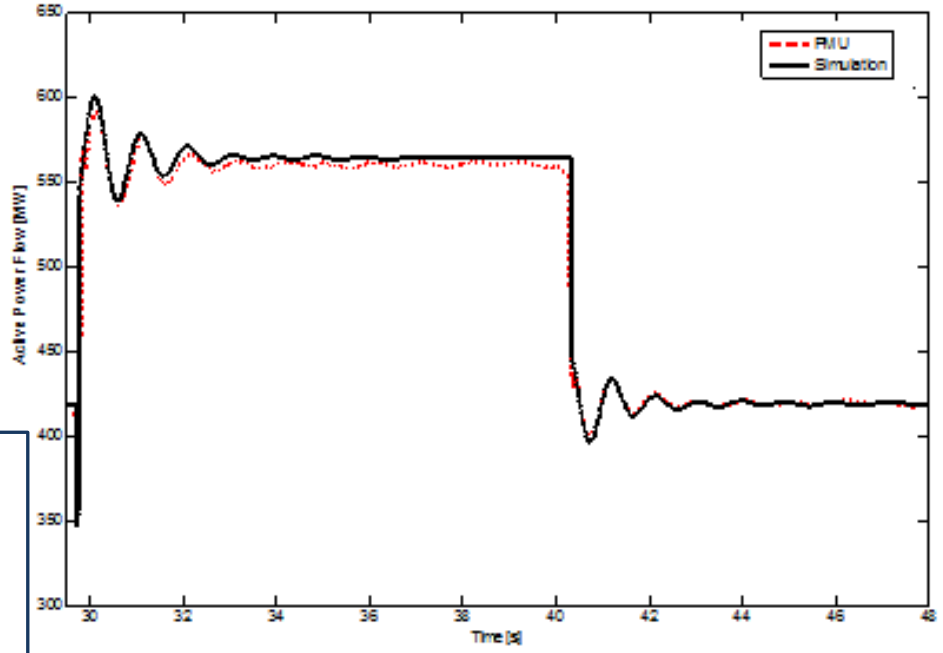
- Two model validation examples done by ISO New England
  - Fault at a 345 kV line followed by successful reclosing
  - Blocking of one pole of an HVDC link due to a fault followed by the blocking of the second pole



MW flow on an AC line

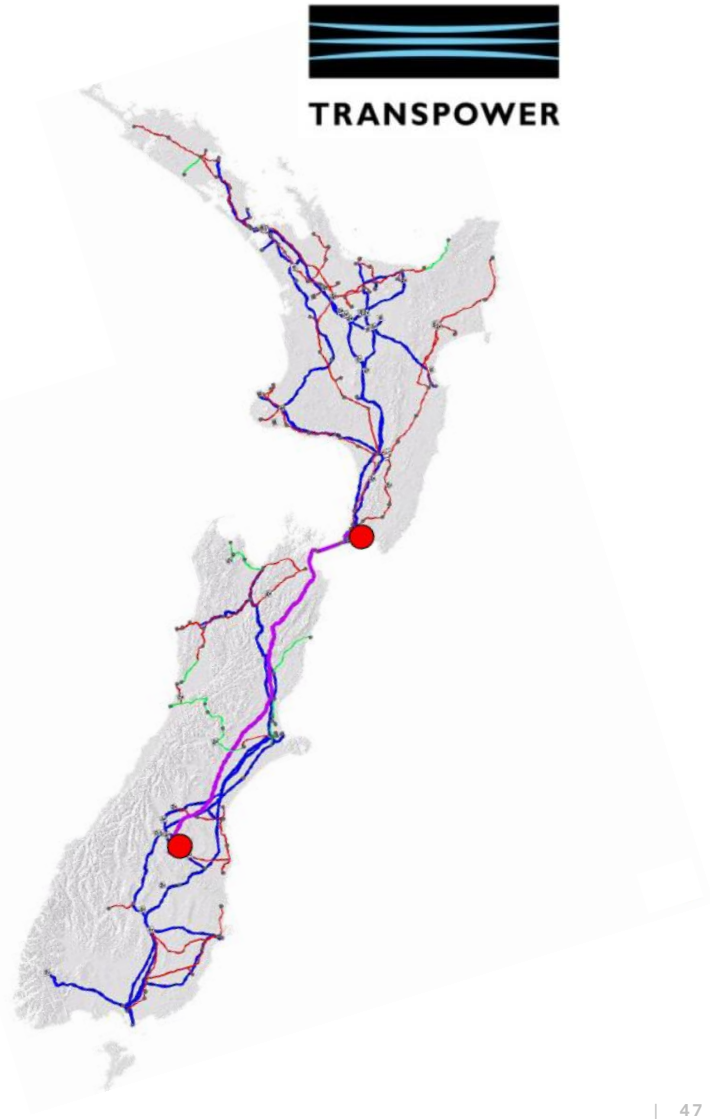
PMU recording

On-line DSA simulation



RESERVE MANAGEMENT @ TRANSPOWER NEW ZEALAND

- Ensuring **acceptable frequency response** after the loss of a large unit or HVDC bi-pole is critical to system operation
  - Appropriate **active power reserve** is required
- A **Reserve Management Tool** (RMT) is used for this purpose in NZ Electricity Market
- RMT runs in the market settlement cycle
  - Full time-domain simulations with reduced network model but detailed dynamic models
  - About 30 sec to solve 6 contingencies
  - Tight integration with market data



IN CASE HELP IS NEEDED…

- **Technical support**
  - For further assistance, contact Powertech at dsainfo@powertechlabs.com

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    Surrey, BC

    Canada, V3W 7R7

- **DSATools™ Web**

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